

Reachability Is Not Selection in Target-Free G_2 Flavor-Frame Dynamics

Finite vacuum, hierarchy, and lepton-link gates

Matthew Connelly
Independent Researcher

14 July 2026

Abstract

We test whether a finite class of target-free exceptional-geometric actions can select a physically meaningful three-family vacuum. Four real three-planes in $\mathbb{R}^7$ define a configuration in $\text{Gr}(3, 7)^4$; a fixed unit vector reduces G_2 to a residual $\text{SU}(3)$, and octonionic tensors generate the scalar action and signed-associator mass maps. In a locked 74-action ensemble built from 21 normalized projector invariants, 68 branches pass a strict stationarity and stability cut and 29 are isolated modulo the known residual orbit. Of the six isolated branches with full-rank mass maps in both sectors, none has two adjacent decade-scale singular-value gaps, while a CKM/PMNS-like matrix is not gauge defined under the independent left-frame basis freedoms.

We then test two remedies. First, the unique primitive chiral loop and the unique lowest-degree exchange-even product within a three-channel complex-loop class are both locally stable, but retain seven and six physical zero modes, respectively. Second, a shared auxiliary $U(3)$ link can be eliminated analytically to produce a nuclear-norm backreaction. In a value-blind but structurally post-exposure extension of the 74 actions, 21 retained branches admit a lepton-angle diagnostic. Sixteen have smooth classical quotient Hessians; all are stable and two are isolated, yet none lies inside all three NuFIT 6.0 three-sigma angle intervals. The smallest best-fit L^1 angle residual is 22.62° and occurs on a non-isolated branch. Direct second-order automatic differentiation reproduces the smooth spectra to a maximum sorted discrepancy of 2.20×10^{-6} . Five further branches are rank deficient, where the nuclear norm is nonsmooth and a classical Hessian is not defined.

These are negative finite gates, not a global no-go theorem. They localize the missing structure: scalar closed-overlap invariants and a singular-value-only link do not provide a nondegenerate family-axis selector. The geometry can represent any unitary mixing matrix, but the tested dynamics does not select one.

1 Introduction

Exceptional groups provide rigid algebraic structures that can sharply constrain model building. The octonions furnish the invariant three-form of G_2 and its dual four-form [1, 2]; fixing a unit imaginary direction leaves an $\text{SU}(3)$ stabilizer and a natural complex structure on the orthogonal six-plane. These ingredients make G_2 geometry a plausible laboratory for asking a precise question: can flavor structure be selected by geometry before observed masses and mixing data are supplied?

That question is stricter than obtaining a good numerical fit. A predictive construction must specify its action without flavor targets, find stationary vacua, establish local stability, quotient all known continuous symmetries, and only then open mass or mixing observables. Failure at any earlier stage prevents a later numerical matrix from being called a prediction. The distinction is especially important for flavor, where unconstrained basis choices can generate apparently realistic mixing matrices without changing any invariant content.

This paper applies that gate discipline to a concrete four-plane G_2 model. It consolidates three computational stages:

Table 1: Evidence labels used throughout the paper.

Label	Meaning
ANALYTIC CONSTRUCTION	Follows from an explicit definition or finite algebraic argument.
VERIFIED COMPUTATION	Reproduced by retained code and machine-readable artifacts, with stated tolerances.
NEGATIVE FINITE GATE	Rules out only the declared action class, starts, operators, and thresholds.
POST-HOC DIAGNOSTIC	Evaluated after structural development; informative but not a held-out prediction.

1. a projector-invariant baseline and a signed-associator mass diagnostic;
2. scalar complex-loop actions designed to correlate all four frames;
3. an auxiliary unitary link whose exact elimination backreacts on the vacuum and permits a shared-space lepton-angle diagnostic.

The central result is negative but specific. Stable vacua are abundant, but predictivity fails in three distinguishable ways: the baseline lacks a gauge-defined relative left basis; scalar loops retain physical flat directions; and the integrated link stabilizes some branches without selecting observed lepton angles.

1.1 Claim-status convention

We use the evidence labels in table 1. They are part of the scientific statement, not editorial decoration.

No result below establishes a Standard Model derivation, a global classification of the invariant ring, or a proof that unobserved stationary branches do not exist.

2 Exceptional-geometric setup

2.1 Octonionic tensors and four three-planes

Let ϕ_{ijk} denote the standard octonionic G_2 three-form on $\mathbb{R}^7$, and let

$$\Phi_{ijkl} = (*\phi)_{ijkl} \tag{1}$$

be its Hodge dual. We fix a unit vector $h = e_7$. The stabilizer of (ϕ, h) is $SU(3)$. On $h^\perp$, define

$$(J_h)_{ij} = \phi_{ijk} h_k, \quad P_h = I - hh^\top. \tag{2}$$

The identities

$$J_h h = 0, \quad J_h^2 = -P_h \tag{3}$$

make J_h a complex structure on the six-plane $h^\perp$.

The dynamical variables are four Stiefel frames

$$X_a \in \mathbb{R}^{7 \times 3}, \quad X_a^\top X_a = I_3, \quad a = 0, 1, 2, 3, \tag{4}$$

interpreted as

$$(X_0, X_1, X_2, X_3) = (L_e, R_e, L_\nu, R_\nu). \tag{5}$$

Their projectors $P_a = X_a X_a^\top$ are invariant under the frame gauges $X_a \mapsto X_a O_a$, $O_a \in O(3)$. The physical configuration space of the baseline action is therefore $\text{Gr}(3, 7)^4$, with 48 local Grassmann coordinates before quotienting the residual $SU(3)$ orbit.

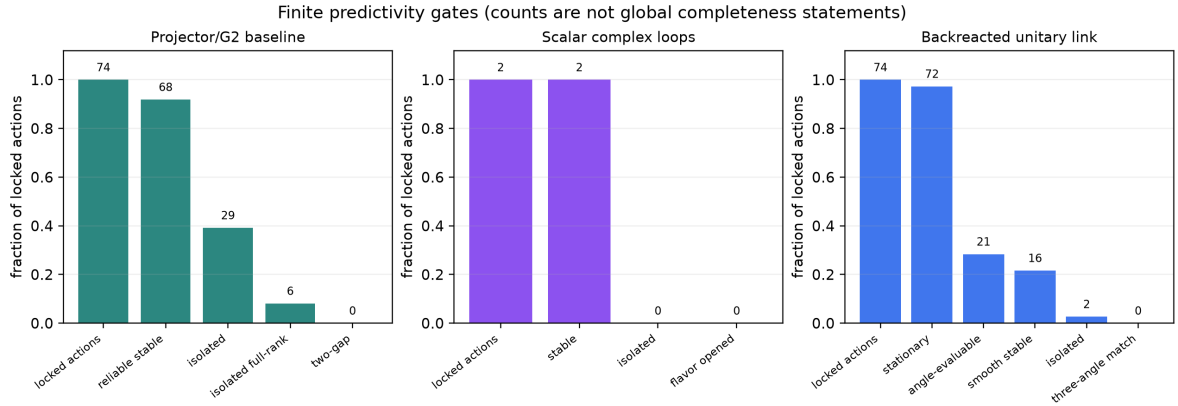


Figure 1: Finite gate counts for the three action classes. Bar heights are normalized by the number of locked actions in each panel, while integers give raw counts. The stages are not global completeness statements and the two-loop panel contains only two symmetry-selected candidate actions.

For later mass diagnostics, the octonionic associator is represented by

$$[e_a, e_b, e_c] = A_{abc}^d e_d, \quad A_{abc}^d = \phi_{abm} \phi_{mc}^d - \phi_{bcm} \phi_{am}^d. \quad (6)$$

All tensor conventions were checked against direct octonion multiplication before optimization.

2.2 Target-free scalar actions

A finite scalar basis was assembled from contractions of P_a , h , ϕ , and Φ . It contains single-frame, pair, triple, and four-frame structures, including traces such as $\text{Tr}(P_a P_b)$ and $\text{Tr}(P_a P_b P_a P_b)$ and tensor contractions of ϕ and Φ with projected indices. Numerical rank-revealing decomposition on a locked Haar sample selected 21 independent feature directions. Each feature was standardized by its sample mean and standard deviation.

The coefficient ensemble contained 74 actions: the positive and negative directions of each primitive feature and pair-sector combinations, together with 32 seeded Rademacher sign vectors normalized in coefficient space. The coefficients and Haar normalization were frozen before any flavor diagnostic. This is a finite computational design, not a basis theorem for all G_2 invariants.

For each action, four generic starts were optimized on the Stiefel product using standard matrix-manifold retractions [3]. A branch was called stationary when the projected gradient norm was below 10^{-5} . The 48-dimensional local Hessian was classified using thresholds

$$\lambda < -10^{-5} \quad \text{negative}, \quad |\lambda| \leq 10^{-5} \quad \text{zero}. \quad (7)$$

The tangent orbit generated by the eight-dimensional residual $\text{SU}(3)$ was constructed explicitly at every vacuum. A stable branch is isolated only when its Hessian kernel contains no directions beyond that orbit.

3 Baseline predictivity gate

3.1 Vacuum stability and isolation

At the strict reliability cut, 68 of the 74 action branches were stationary and stable, and 29 were isolated modulo the explicitly constructed residual orbit. These counts differ slightly from a raw Hessian-only classification because the strict cut also excludes a numerically unreliable stationary representative. The search used the best retained representative among four starts and therefore does not certify that every stationary branch has been found.

3.2 Signed-associator mass map

Flavor was opened only after stationarity, stability, and degeneracy classification. A composite direction was declared by

$$V = \text{top eigenvector of } P_h \left(\sum_{a=0}^3 P_a \right) P_h, \quad (8)$$

provided that the top spectral gap exceeded 10^{-8} . For a left/right pair (L, R) , the signed-associator matrix was

$$Y_{ij} = \langle [L_i, V, h], R_j \rangle. \quad (9)$$

Its singular values are invariant under the independent frame gauges. Thirty-six reliable stable branches had a usable composite direction; 11 of those were isolated, and six were isolated with full-rank maps in both sectors.

An earlier endpoint-span count can mistake a single large gap for a three-rung hierarchy. We therefore required both adjacent singular-value gaps to exceed one decade in each sector. None of the six isolated, full-rank two-sector branches passed. This is the first negative finite gate:

NEGATIVE FINITE GATE. No isolated, full-rank, two-gap hierarchy occurs in the tested 74-action ensemble with the declared composite- V signed-associator map.

It is not a no-go theorem for another mass operator or for the full invariant ring.

3.3 Why baseline mixing is undefined

Let Y_e and Y_ν be the two real 3×3 matrices obtained from eq. (9). A formal mixing proxy formed from their left singular vectors changes under the independent frame transformations

$$L_e \mapsto L_e O_e, \quad L_\nu \mapsto L_\nu O_\nu. \quad (10)$$

The singular values are unchanged, but the relative left basis is arbitrary. Consequently a CKM/PMNS-like matrix is not a physical observable of the baseline action. This is a structural obstruction, not a failure to choose a preferred singular-value convention.

4 Complex overlap and scalar loop actions

The exceptional complex structure in eq. (2) defines complex overlaps

$$K_{ab} = X_a^\top (P_h + iJ_h) X_b. \quad (11)$$

Up to a factor of two, $P_h + iJ_h$ is a rank-three complex projector on the complexified six-plane. The frame transformations act as

$$K_{ab} \mapsto O_a^\top K_{ab} O_b, \quad (12)$$

so traces around closed frame cycles are gauge invariant.

4.1 Exact reachability does not imply dynamical selection

The overlap geometry is not too small to represent realistic mixing. Choose an orthonormal Lagrangian frame $P = (p_1, p_2, p_3) \subset h^\perp$ satisfying

$$P^\top J_h P = 0, \quad (13)$$

and set $Q = -J_h P$. The six columns of (P, Q) are orthonormal. For any target $U \in U(3)$, define

$$L_e = P, \quad L_\nu = P \Re U + Q \Im U. \quad (14)$$

Table 2: Discrete characters of the three primitive loop channels.

Channel	sector exchange	left/right exchange	CP
I_1	+1	-1	-1
I_2	-1	-1	-1
I_3	-1	+1	-1

Then L_e and L_ν are real orthonormal frames and direct substitution gives

$$L_e^\top (P_h + iJ_h)L_\nu = U. \quad (15)$$

Thus the polar-link map is surjective onto $U(3)$. A stress test over 1000 seeded Haar-random unitaries gave a worst polar residual of 1.34×10^{-15} . This is an analytic representability statement with a numerical implementation check, not a vacuum-selection result. Indeed, on the unitary image of eq. (15), the nuclear norm is the constant $\|U\|_* = 3$ and cannot distinguish orientations.

4.2 Three primitive oriented cycles

There are three inequivalent oriented four-cycle pseudoscalars in the declared class:

$$\begin{aligned} I_1 &= \Im \text{Tr}(K_{01}K_{12}K_{23}K_{30}), \\ I_2 &= \Im \text{Tr}(K_{01}K_{13}K_{32}K_{20}), \\ I_3 &= \Im \text{Tr}(K_{02}K_{21}K_{13}K_{30}). \end{aligned} \quad (16)$$

Their characters are shown in table 2. Complex conjugation reverses all three signs.

Requiring sector-even, left/right-odd, and CP-odd character selects the primitive action

$$V_\chi = -\chi \frac{I_1}{\sigma_1}, \quad \chi = +1, \quad (17)$$

within this finite class. The opposite sign is the CP-conjugate convention. After the primitive candidate failed isolation, the lowest-degree monomial containing all three characters and even under both exchanges was

$$V_3 = -\chi \left(\frac{I_1}{\sigma_1} \right) \left(\frac{I_2}{\sigma_2} \right) \left(\frac{I_3}{\sigma_3} \right), \quad (18)$$

where the locked Haar standard deviations are

$$(\sigma_1, \sigma_2, \sigma_3) = (0.2786481, 0.2812664, 0.2736715). \quad (19)$$

“Unique” here means unique under the stated character and degree restrictions, not unique in the complete invariant ring.

4.3 Quotient-Hessian result

Each action was solved from 96 generic starts. Every retained start approached the same numerical energy for its action, but this empirical convergence is not a proof of global minimality. The local results are in table 3.

Both stationary representatives are stable within tolerance. Neither is isolated after subtracting the rank-eight residual $SU(3)$ orbit. The three-channel product reduces the number of extra physical zero modes from seven to six, but does not eliminate the continuous ambiguity. Flavor was therefore not opened on either vacuum.

Table 3: Quotient-isolation gates for the two scalar complex-loop actions. Eigenvalue counts use the tolerances in eq. (7).

Action	$\ \nabla V\ $	negative	zero	positive	extra physical zero
V_χ	3.07×10^{-14}	0	15	33	7
V_3	1.50×10^{-11}	0	14	34	6

5 Auxiliary unitary link and exact backreaction

5.1 Shared left-generation space

To compare the charged-lepton and neutrino left spaces, introduce a bifundamental link $\Sigma \in U(3)$ and define

$$K \equiv K_{02} = L_e^\top (P_h + iJ_h)L_\nu. \quad (20)$$

Under full-rank frame changes,

$$K \mapsto O_e^\top K O_\nu, \quad \Sigma \mapsto O_e^\top \Sigma O_\nu. \quad (21)$$

The auxiliary interaction is

$$V_{\text{aux}}(\Sigma, K) = -\Re \text{Tr}(\Sigma^\dagger K). \quad (22)$$

If $K = U \text{diag}(s_1, s_2, s_3)V^\dagger$ is a singular-value decomposition, the minimum is attained by the polar factor $\Sigma = UV^\dagger$ and

$$\min_{\Sigma \in U(3)} V_{\text{aux}} = -\sum_i s_i = -\|K\|_*. \quad (23)$$

Thus the link can be integrated out exactly. The result is invariant under the independent frame gauges, but it depends only on the singular geometry of K . For rank-deficient K , the minimum value remains well defined but the unitary polar completion is not unique; mixing angles obtained from a particular null-space completion are therefore not physical diagnostics.

For a locked standardized baseline action V_c , the tested backreacted action was

$$V_{\text{tot}} = V_c - \frac{\|K\|_* - \mu_K}{\sigma_K}, \quad \mu_K = 2.1768243, \quad \sigma_K = 0.2519311. \quad (24)$$

The coefficient -1 is a standardized modeling convention. It is not derived from a coupling unification or a representation-theoretic normalization. No PMNS angle or mass enters eq. (24), but the link architecture was introduced after the baseline mixing obstruction was known. We therefore call the extension *value-blind but structurally post-exposure*, rather than a fully held-out prediction.

5.2 Lepton-angle diagnostic

For each sector, define the positive left operator

$$J_f = Y_f Y_f^\dagger. \quad (25)$$

The neutrino operator is transported into the charged-lepton left space with Σ . If U_e and U_ν diagonalize the corresponding operators in their native spaces, the shared-space mixing matrix is

$$U_\ell = U_e^\dagger \Sigma U_\nu. \quad (26)$$

Its absolute values and rephasing invariants are physical under the frame gauges; raw matrix entries remain subject to conventional phases and permutations. The three standard angles are extracted from $|U_{\ell,13}|$, $|U_{\ell,12}|$, and $|U_{\ell,23}|$ in a fixed mass ordering, while the Jarlskog invariant is

$$J = \Im(U_{11}U_{22}U_{12}^*U_{21}^*) \quad (27)$$

Table 4: Backreacted lepton-angle and quotient-Hessian gate. “N/A” marks rank-deficient branches at which the nuclear norm is nonsmooth.

Gate quantity	Count
locked actions	74
retained stationary actions	72
angle-evaluable branches	21
smooth classical Hessians	16
rank-deficient/nonsmooth	5
quotient-stable smooth branches	16
quotient-isolated smooth branches	2
three-angle-compatible and stable	0

following the usual rephasing-invariant construction [4, 5].

The numerical comparison uses the normal-ordering NuFIT 6.0 analysis with atmospheric information [6]. The best-fit angles are

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (33.68^\circ, 43.3^\circ, 8.56^\circ), \quad (28)$$

with three-sigma intervals

$$[31.63, 35.95]^\circ, \quad [41.3, 49.9]^\circ, \quad [8.19, 8.89]^\circ. \quad (29)$$

Although the benchmark also records δ_{CP} , the present gate scores only the three angles. We therefore use “three-angle compatible,” not “PMNS compatible,” for a pass.

6 Backreacted quotient-Hessian gate

6.1 Smooth, saturated, and nonsmooth branches

The 74 backreacted actions were solved from four generic starts each; 72 had a retained stationary representative. Requiring a usable composite direction and nondegenerate two-sector mass spectra left 21 angle-evaluable branches. Their Hessian treatment splits into three cases:

1. five unitary-saturated branches, evaluated with the exact second-order germ of the nuclear norm;
2. eleven full-rank, nonunitary branches, evaluated by centered finite differences of the true-SVD gradient and independently by direct second-order automatic differentiation through the SVD;
3. five rank-deficient branches, where the nuclear norm is nonsmooth and no classical Hessian is assigned.

For the smooth branches, the residual $\text{SU}(3)$ orbit was projected out before eigendecomposition. The independent verifier reproduced all 16 smooth classifications and the five nonsmooth identifications. The maximum discrepancy between sorted finite-difference and direct-autograd spectra was 2.20×10^{-6} , below the declared 2×10^{-5} comparison tolerance.

6.2 Angle and isolation results

Every smooth branch is stable within tolerance, but only two are isolated. Those two isolated solutions have angle triples

$$\begin{aligned} (5.105^\circ, 3.263^\circ, 0.300^\circ), & \quad \Delta_1 = 76.87^\circ, \\ (0.00058^\circ, 1.774^\circ, 0.00078^\circ), & \quad \Delta_1 = 83.77^\circ, \end{aligned} \quad (30)$$

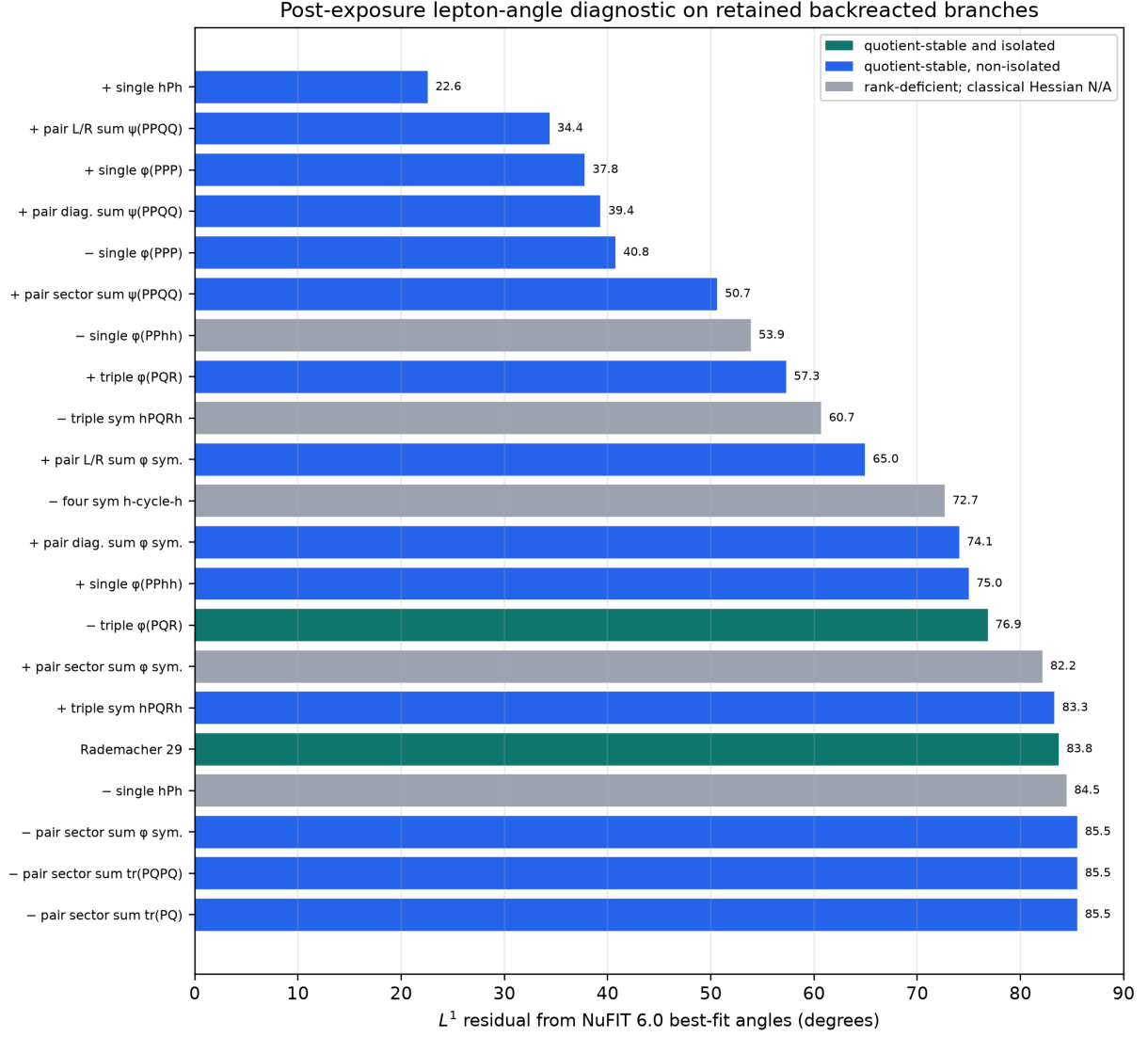


Figure 2: Post-exposure three-angle diagnostics, sorted by L^1 residual from the NuFIT 6.0 best fit. Blue bars are stable but non-isolated smooth branches and green bars are stable and isolated. Gray bars are rank-deficient branches: the displayed values record the SVD convention used in the retained computation, but a classical Hessian is not defined and the unitary null-space completion is not unique. The CP phase is not scored.

where

$$\Delta_1 = \sum_{ij \in \{12, 23, 13\}} |\theta_{ij} - \theta_{ij}^{\text{NuFIT best}}|. \quad (31)$$

The closest retained branch has

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (16.889^\circ, 42.683^\circ, 13.773^\circ), \quad \Delta_1 = 22.622^\circ, \quad (32)$$

with $J = 0.01065$. It is stationary and quotient-stable, but has 22 quotient zero modes and is therefore not isolated.

The decisive result is therefore

NEGATIVE FINITE GATE WITH INDEPENDENTLY VERIFIED HESSIAN SPECTRA. Among the 16 smooth angle-evaluable branches found by the four-start 74-action solve, no quotient-stable branch lies inside all three NuFIT 6.0 three-sigma angle intervals. Five additional rank-deficient branches are nonsmooth and are not assigned classical stability.

The statement does not exclude an unobserved branch, another relative link coefficient, a different mass map, or an action outside the locked class.

7 Structural interpretation

The failures are informative because they occur at different gates.

First, projector-only scalar actions can isolate three-planes while leaving their internal left bases independent. Mass singular values survive this gauge freedom; mixing does not. Second, the complex loops in eq. (16) correlate all four frames, but scalar traces still leave continuous physical moduli. Third, integrating out Σ produces only $\|K\|_*$. At unitary saturation, all three singular values of K are fixed and the nuclear-norm term is constant over the reachable orientation family. It can stabilize singular geometry without selecting the observed unitary orientation.

These observations point to a more specific missing object than simply “more terms.” A viable extension requires a matrix-valued family selector, say T_L , with a nondegenerate spectrum in the shared left-handed generation space. Schematically one may consider

$$V_{\text{family}} = \sum_{f=e,\nu} \|T_L K_{L_f R_f} - K_{L_f R_f} T_{R_f}\|_F^2 - \chi \Im \text{Tr}(T_L K_{01} K_{12} K_{23} K_{30}). \quad (33)$$

Equation (33) is a search direction, not a locked model. To become predictive, the representation of T_L , its self-potential, the right-handed tensors, and every relative coefficient must be derived without flavor targets. Its quotient Hessian must then leave only gauge or explicitly classified discrete degeneracies.

A concrete physical interpretation would be a dynamical family-order parameter shared by the charged-lepton and neutrino Yukawa sectors. In a four-dimensional effective theory this could arise from an adjoint or higher-rank condensate; in a condensed-matter analogue it resembles a nondegenerate mode-splitting tensor that selects axes inside a degenerate multiplet. The present scalar actions lack precisely that axis-resolving information.

8 Limits of the calculation

The following boundaries are essential.

1. **Finite action class.** The 21 features are numerically independent on a locked sample, but they do not constitute a theorem about the full invariant ring.
2. **Finite starts.** Four generic starts were used per baseline and backreacted action. The loop candidates used 96 starts. Neither protocol proves global completeness.
3. **Local Hessians.** Stability is local and tolerance dependent. Rank-deficient nuclear-norm branches are nonsmooth and are not silently classified as stable or unstable.
4. **Model-dependent mass map.** The signed-associator map and composite direction in eqs. (8) and (9) are declared diagnostics, not derived Standard Model Yukawa operators.
5. **Post-exposure lepton test.** No observed angle enters the backreacted objective, but the shared-link architecture was introduced to repair a known mixing obstruction. The angle comparison is exploratory rather than a pristine blind prediction.
6. **Angle-only score.** The CP phase and oscillation mass splittings are not included in the compatibility gate.
7. **No zero-parameter claim.** Haar standardization fixes numerical scales for comparison, not physical coupling constants. In particular, the relative coefficient of the nuclear term is assumed.

8. **No ultraviolet completion.** Connecting this local four-plane model to an exceptional gauge theory such as E_6 would require an explicit representation-level map and a derived neutral-fermion operator; neither is supplied here.

9 Conclusion

The four-plane G_2 model passes several nontrivial mathematical and numerical checks: its tensor kernel is self-consistent, many locked scalar actions possess stable stationary vacua, some are isolated modulo the residual $SU(3)$, closed complex overlaps have the required gauge and discrete characters, and the integrated unitary link admits independent second-order verification. Those successes do not produce a flavor prediction.

The baseline action class fails to give both a three-rung hierarchy and a gauge-defined mixing matrix. The two symmetry-selected scalar loop actions are stable but non-isolated. The nuclear-norm backreaction produces 16 smooth stable lepton-angle branches, including two isolated branches, but none satisfies all three NuFIT 6.0 three-sigma angle intervals. The closest branch is non-isolated and remains 22.62° away in the angle-only L^1 diagnostic.

The appropriate conclusion is not that exceptional flavor geometry is impossible. It is that this minimal scalar architecture lacks a nondegenerate family-axis selector. A next model should derive such a matrix-valued order parameter first, pass stationarity and quotient-isolation gates without flavor data, and only then evaluate masses and mixing.

A Numerical verification and artifact ledger

All optimization and Hessian calculations used double precision. The retained workflow separates action locking, vacuum solving, quotient-Hessian classification, and flavor scoring. The backreacted nonunitary Hessians were recomputed by direct second-order automatic differentiation through the true singular-value decomposition; the saturated unitary germ was verified separately; and rank-deficient points were recorded as nonsmooth rather than regularized into a classical Hessian.

The source distribution contains a machine-readable claim ledger with SHA-256 hashes for the authoritative JSON artifacts. The principal artifact families are:

- target-free action basis, vacuum, stability, post-stability flavor, and shape-verifier results;
- exact polar-link reachability construction and 1000-case unitary stress test;
- primitive and three-channel loop action locks, vacuum solves, and quotient-Hessian gates;
- backreacted link action lock, vacuum solve, angle-evaluable quotient-Hessian gate, and independent autograd verification;

The claim ledger records the exact boundary and source hashes of each finite gate.

References

- [1] John C. Baez. The octonions. *Bulletin of the American Mathematical Society*, 39(2):145–205, 2002.
- [2] Robert L. Bryant. Metrics with exceptional holonomy. *Annals of Mathematics*, 126(3):525–576, 1987.
- [3] P.-A. Absil, Robert Mahony, and Rodolphe Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, Princeton, NJ, 2008.

- [4] Makoto Kobayashi and Toshihide Maskawa. CP-violation in the renormalizable theory of weak interaction. *Progress of Theoretical Physics*, 49(2):652–657, 1973.
- [5] Cecilia Jarlskog. Commutator of the quark mass matrices in the standard electroweak model and a measure of maximal CP nonconservation. *Physical Review Letters*, 55:1039–1042, 1985.
- [6] Ivan Esteban, M. C. Gonzalez-Garcia, Michele Maltoni, Ivan Martinez-Soler, João Paulo Pinheiro, and Thomas Schwetz. NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations. *Journal of High Energy Physics*, 12:216, 2024.